

IR Remote-Controlled RGB BULB

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Infrared (IR) remote-controlled RGB bulbs available in the market are extremely expensive. Here is a simple, easy-to-build and inexpensive circuit built using Arduino Uno R3. The project demonstrates how to turn on/off or vary the colours of an RGB LED bulb using an IR remote (used for televisions, air-conditioners, set-top boxes and so on). The author's prototype is shown in Fig. 1.

Arduino varies the intensity of RGB LED by generating pulse-width modulation (PWM) signals through its digital output pins. The LED has three internal LEDs: red, green and blue. Arduino varies the intensity of each internal LED and generates different mixtures of colours such as cyan, pink, yellow and magenta.

Primary colours are red, green and blue. The concept of mixing these is shown in Fig. 2. Additive mixing of red and green produces shades of yellow, orange or brown. Mixing green and blue produces shades of cyan, and mixing red and blue produces shades of purple (including magenta).

Circuit and working

The circuit of IR remote-controlled RGB bulb is shown in Fig. 3. It is built around Arduino Uno board, common-anode RGB LED (RGB1), incandescent bulb's glass envelope, 9V battery, IR receiver module TSOP1738 (or equivalent) and IR remote (Vire).

Sensor TSOP1738 has three terminals: Gnd (pin 1), Vcc (pin 2) and output (pin 3). Its Vcc terminal is connected to 5V and Gnd to ground pins of Arduino, respectively. Sensor output is connected to digital input pin 2 of Arduino. RGB1 common-anode terminal is connected to 5V supply through current-limiting resistor R1 (100-ohm). Internal LED terminals are connected to digital pins 11, 10 and 9 of Arduino, respectively.

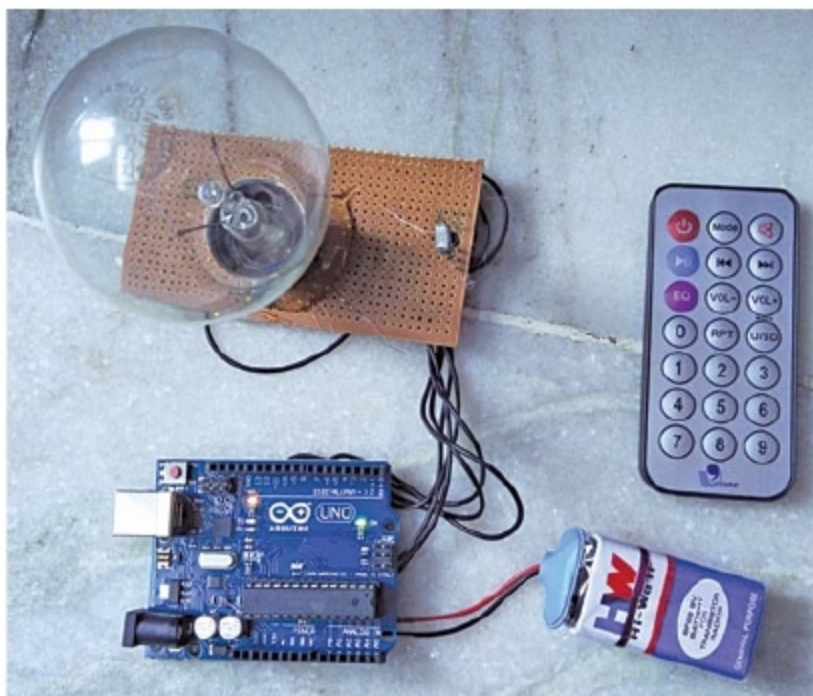


Fig. 1: Author's prototype



Fig. 2: Primary colour mixing

IR remote configuration. Select any eight buttons of IR remote for different functions, to perform the following actions:

1. Turn on/off RGB LED
2. Increase/decrease red colour intensity
3. Increase/decrease green colour intensity
4. Increase/decrease blue colour intensity

MP3 Vire IR remote has many buttons, including digit buttons (0 to 9), volume control, forward/reverse, play/pause and so on. The eight buttons and their functions used in this project are given in Table I.



Fig. 4: IR sensor TSOP1738

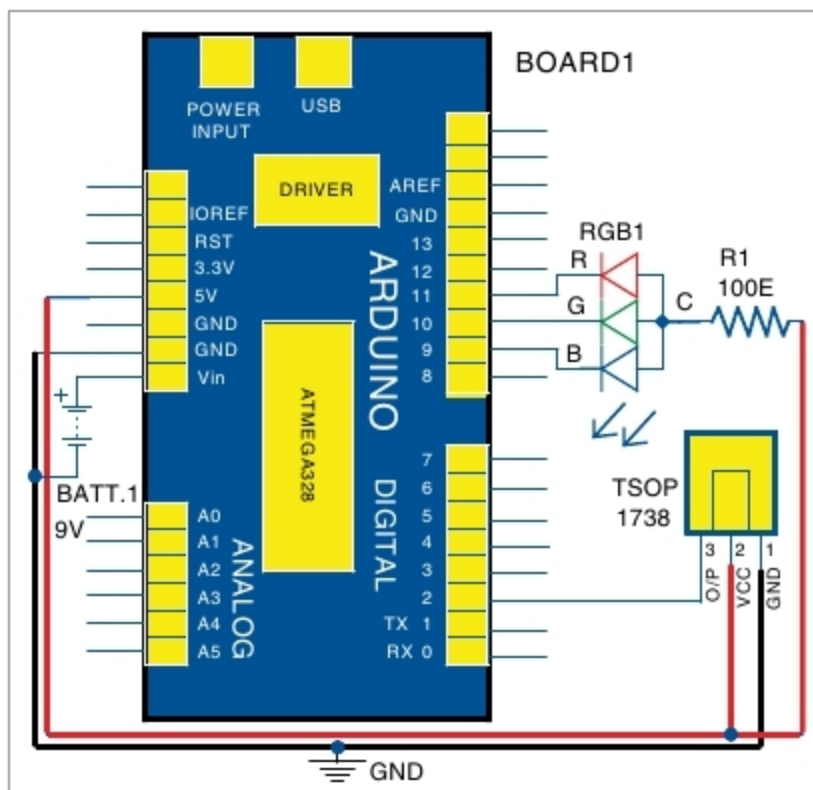


Fig. 3: Circuit diagram of IR remote-controlled RGB bulb

Next, decode the codes for these buttons. When a button is pressed, the corresponding code transmits through its internal IR LED of the remote. To decode this code, IR remote functions and IR remote library for Arduino are used. These libraries are readily available on the Internet. Download the libraries and include in Arduino Libraries folder.

Compile the code/sketch (rgb_vire_remote.ino) and upload